

VALQUO_MASTER_AUDIT_4.md — the frontier audit

Commissioned by `PROMPT_audit4_master.md`. **Executed 2026-08-19**. **Cold auditor, strictly read-only**. No file under `valuation/`, `scripts/`, `tests/`, `data/` or `.github/` was changed. No backtest was run. **Zero trials charged** — every measurement in this document is a fact about what is on disk or what the code does, in the class of `S25`, the options re-open list and `HANDOFF_data_mining.md`. Nothing is adopted, nothing is re-opened, and no verdict moves.

Items are IDed `MB1 ... MB42` (MA is taken by audit #3). Grouped by mandate, EV-ranked within group. **42 items; total trial cost if every one were run is 32, of which 16 is MB2's grid, which I recommend against — so the whole of the rest of this audit costs 16 trials.**

0. THE BOARD WAS QUIET, AND ONE ROW SAYS OTHERWISE FOR A REASON THAT IS ITSELF A FINDING

The commission forbids starting on a moving target. Checked four ways rather than one:

check	result
working tree	clean but for the untracked commission itself
origin/worktree- * not an ancestor of origin/main	0
remote refs not merged	2, both <code>rescue/*</code> , both relics of the 2026-08-10 PT-WRITER incident (MB35)
ledger status cells containing <code>IN PROGRESS</code>	1 — B13, whose own cell reads <i>"PARTIAL - BLOCKED ON DATA, NOT IN PROGRESS"</i>
ledger status cells containing <code>INPROGRESS</code>	1 — D11, and it is stale

D11's row is dated 2026-08-17 and reads `INPROGRESS`. Its own handoff, written the next day and committed at `d6c943c` ("The harvest is closed"), opens: ***"Status: HARVEST CLOSED 2026-08-18. THE MINER IS IDLE AND THE SUBSCRIPTION CAN LAPSE ON SCHEDULE."*** All five tiers complete, 2,850 units, zero failed, frozen and hash-verified.

So the lane is idle and its board row says it is working. That is mandate 7's thesis reproduced on the first check of the audit, and it is why MB27 is ranked where it is. The board is quiet; I proceeded.

One more thing that had to be re-derived rather than trusted. The commission states equity `N 232+`, options `297+`. Measured live from `research_log.detail()` at session start:

```
trials_logged 549   by_domain {'equity': 234, 'options': 300, 'unified': 0, 'infra': 15}
rows_counted 139   rows_fixed_not_counted 65   trials_domain_unresolved 0
```

Every bar in this document is quoted at equity `N = 234` (hurdle `3.3031261300040304`) or options `N = 300` (hurdle `3.3775086897463940`). Where I quote a bar at a different `N`, I say so.

1. THE HONEST ONE PAGE

The three most valuable frontiers

FIRST — `S22`'s term-structure claim has never faced the one null that predicts its own shape, and I measured the gap rather than asserting it (MB21). Boudoukh–Richardson–Whitelaw (RFS 2008) is that with a *persistent* regressor and *overlapping* long-horizon returns, R^2 and long-horizon β rise **mechanically under the no-predictability null**. `S22`'s per-horizon null is `fixed_weights_null`, built from `placebo_panel`, which permutes signal rows **within each rebalance date** — so the placebo signal has, by construction, no persistence at all. Measured read-only on `panel_corrected_69d.pkl`: the **real** composite's per-name lag-1 rank autocorrelation across rebalances is **0.5802**, still **0.4099 at lag 8 (two years)**; the **placebo's is 0.000** ($-0.0011 / +0.0010 / +0.0005$ across three seeds). The most persistent theme is `size` at **0.9915** — and `x3` measured that `size` carries the composite's entire statistical significance. So every horizon beyond `H=63` in `S22` is scored against a null that cannot generate the artifact it most needs to exclude, and the error grows with horizon, which is precisely the axis `S22`'s headline lives on. **This is product-consequential**: `S22-DISPLAY` ships *"still ahead by about 5.1% annualized two years later"* and the rank-IC-rises-with-horizon corroboration on the hot-list card, the name row and `/methodology`. Cost to resolve: **1**

infra trial, no new data, one session.

SECOND — 2.7 million alternative contracts have been asked exactly one question, and the question that decomposes R2 has never been asked at all (MB1). The harvest census pinned ~636 alternatives per entry date and 99.9% holding-period chain coverage; 021-D2 then spent the one authorised same-hypothesis re-open on them and came back IMMATERIAL, with both statistics' CI95 inside the bar. What no prior study could do — because every prior study scored the ONE contract the book held — is compare **menus** rather than picks: the distribution of outcomes over the whole in-band alternative set on an alert day, against the same menu on random days. That separates "*the day was bad*" from "*the contract was bad*", which R2 and 013 between them cannot. 013 already measured the loss to be **entirely a within-bin rate effect** (−4.23pp to −5.79pp against a −5.0640pp total), which makes the expected answer "*it is timing*" — and a pre-committed confirmation of that **closes contract selection permanently for 2 trials**, which is the cheapest closure available anywhere in the options book.

THIRD — the incremental-IC gate that this commission itself prescribes for every proposed cross is defective on this panel, and I reproduced the defect independently (MB7). MA58-SEAS found it on 2026-08-18; measured here from the panel directly: complete-case residualisation on the seven weighted incumbents leaves **49 of 69 dates**, first date **2014-01-17**, because institutional has coverage **0.7172** and starts there. Rows fall from **81.2% to 58.3%**. Dropping institutional from the residualisation basis restores **69 of 69**. So **every** mandate-2 and mandate-4 proposal in this document is silently a post-2014 test until the gate is re-specified — and the re-specification is free. It is the highest-EV item in the audit per unit of cost, because it costs nothing and it gates six other items.

The state of the project, in one paragraph

Valquo has done the expensive thing properly: 549 pre-registered trials across two books, an adversarially calibrated placebo floor, a trial denominator that feeds its own Deflated Sharpe, a public research record, and one adoption. The equity composite clears its own calibrated bars (top-decile alpha HAC t 4.3762 against a floor of 2.0540; long-short HAC t 2.6199 against 2.2837) and **fails** the bar derived from counting its own trials (3.3031 at $N = 234$, shortfall 0.6832) — a tension the project states honestly and correctly declines to resolve. The options book is 300 trials with zero entry edges and several formally closed families. What the record now shows, and what audit #4 is for, is that the *marginal* return to further searching this panel has gone close to zero: the arithmetic in MB13 shows that a regime contrast on the panel's single biggest open finding would need **34 years of data on each side** and there are 17.3 years in total, and the same shape recurs everywhere — MA58-SEAS, V6, S19, MA31/MA32 and U2 all returned nulls that the projects' own power rules could not interpret. **The frontier is therefore not another hypothesis. It is (a) instruments — a persistence-preserving null, a repaired incremental-IC gate, a ported effective-n bar — and (b) objects the project owns and has never read: 2.7 M alternative contracts, a 14-venue tick tape, and a banked reverse-DCF panel with a point-in-time expectations gap in it.** Three of the four cheapest items in this audit cost zero or one trial and two of them are corrections rather than searches.

What I could not check

• **The three freezes are on D: and I did not open them.** D:\thetadata\freeze_options_2026-08-17 (12,302 files / 26.98 GB), freeze_rawpull_2026-08-18 (2,850 units / 17.69 GB) and the harvest census JSON are cited from HANDOFF_data_mining.md. Every alternatives figure in mandate 1 is **that handoff's**, cross-checked against the options re-open list where the two overlap (the 699.6 mean slice reproduces across two independent passes) but not re-measured by me.

• data/free_analysis/MA28_CARD.json **is not on the primary checkout's disk**, though valuation/web/accounting_risk.py:120 names it as the shipped card's ARTIFACT. Its figures are pinned as literals in that module and its FORMULA_SOURCE script exists, so nothing is lost — but the provenance pointer a reader is invited to follow dangles. RUN_RULES rule 9's last mile. Sibling artifacts from the same era (S10_ACCOUNTING.json, MA58_SEASONALITY.json, 021D2_ALT_PNL.json) are all present, so this is specific rather than systemic.

• **The live service.** I did not authenticate to any surface, so every claim about what renders is read from the module that owns the copy, never from the site.

• ext_condition1-4 **in the tick cache read 255 on all rows of all three sampled payloads.** That is a sampled observation, not a census; four of the cache's 22 columns may carry no information and I did not prove it across 3,884 units.

• **TIDEMARK's numbers** are read from its generated documents (POWER_GATE.md is machine-generated and says so). I re-ran nothing there.

MANDATE 1 — THE OPTIONS EDGE, RELOCATED

EV-ranked. The dead questions are not re-litigated; R2, U1, 06, 013 and P1S0 are cited against the proposals rather than around them.

MB1 — The alternatives menu, and the selection-vs-timing decomposition it uniquely permits

EV: HIGH (a cheap permanent closure). Trials: 2, options (N 300 → 302). Prior: ~20% that selection carries any of the loss.

Mechanism. Every options study this project has run scored the ONE contract the book held. R2 therefore measures a compound: the alert chose a *day* and a *rule* chose a *contract*, and the -5.0640pp gap cannot be attributed to either. The census pins **2,713,919 alternative contracts, median 636 per entry date across 8 expirations and 61 strikes, on 3,885 of 3,885 entry dates**, with a frozen full chain on every session in the Tier-C/D/E pull. That makes a *distributional* object available for the first time: the outcome distribution of the **whole in-band menu** on an alert day, versus the same menu construction on matched random days.

If the alert-day menu distribution and the random-day menu distribution coincide, the loss is **100% timing** and contract selection is irrelevant — which closes contract selection for good. If the alert-day menu is systematically worse, the alert is picking bad *chains*, not bad *days*, and 013's within-bin finding needs re-reading.

Why this is not 06 in costume. 06 varied four cheapness rules and compared *picks*; its decisive finding was that every rule silently **changed the delta** (mean absolute delta gap up to 0.310), so it never held exposure fixed. This scores the **entire menu, bucketed by delta**, so delta is a reported axis rather than an uncontrolled by-product — which is exactly the remedy MA54-4 prescribed and 06 could not apply. It is also not 021-D2: that priced **one** alternative per entry (the dividend-corrected pricer's different pick on 179 entries) and returned IMMATERIAL; this forms a distribution over ~636.

Register. Pre-commit, before any outcome is read: the menu definition (moneyness band and DTE band, taken verbatim from the shipped `pick_contract` prefilter so it is the engine's own menu); the primary statistic (menu **median** holding-period return, with menu p75 as the declared secondary); the comparator (5-seed split-clean matched random entry, R2's own control books); the both-halves rule; and the delta-bucket reporting grid.

Kill condition. If the alert-day and random-day menu medians differ by **less than 1.0pp in either half**, contract selection is declared irrelevant, the register says so in advance, and **no further contract-selection register may be opened on this book**. A pass does not license a trade — 011 binds (see MB2).

Inherited caveats. U1-SPLIT's rule travels: strikes are as-traded, `raw_close` for anything touching a strike and `close` only for a return. MA31's pair-availability warning travels: a two-sided usable quote on a menu is rarer than on a single pick, so the effective menu will be smaller than 636.

MB2 — The DTE × delta grid, designed honestly — and the design is a recommendation against running it

EV: MEDIUM (the design is the deliverable). Trials if run: 16, options (N 300 → 316; hurdle 3.3775 → 3.4034). Prior: ~7% that any cell is ACTIONABLE.

Don has pre-committed interest in 60–90 DTE. The commission asks for the full-grid register with multiplicity priced in advance. Here it is, and then here is why I would not spend it.

The grid. DTE {30, 60, 90, 180} × delta {0.15, 0.35, 0.60, 0.85} = **16 cells**. All sixteen booked as trials in the register **before** the run and **all sixteen reported**, pass or fail — the $n < k$ grid convention the research log already carries.

The survivorship rule, stated in advance. Contracts are chosen from the frozen menu on the entry date and settled from the frozen forward path, hold-to-expiry. A cell with **fewer than 30 closed legs in either half is UNDERPOWERED, never null** — the V6-B M2 / 026 floor precedent. Cells are not merged to reach the floor.

What would make a cell ACTIONABLE rather than merely significant — and this is the leg that binds. 011 measured that a book with **+3.27%/trade positive expectancy ends at \$37,059 from \$50,000** at a concurrency cap of 10, because the edge lives in the crowded weeks (expectancy -4.51% in quiet weeks, **+14.28%** in weeks above

the 90th percentile, 51.5% of trades in weeks of more than 10 alerts) and a cap refuses exactly those. So a cell is ACTIONABLE only if **all four** hold: (1) expectancy positive; (2) in both halves; (3) clearing its own within-cell permutation p95; and (4) **terminal equity above starting equity at 011's cap-10, \$50k configuration**. Leg 4 is the one that will fail, and a register that omits it produces a significant cell nobody can trade — which is the exact failure 011 exists to prevent.

Why I recommend against it. Three separate measurements already price the corners. The deep-ITM long-tenor corner is DEEPTM-FIN: **executable financing rf + 342 bps, all-in 701.87 bps/yr**, and the roll is half of it and cannot be avoided. The 35-delta short-tenor corner is the banked book, and R2 is standing. The whole family that would express any winning cell was closed by P1S0 on a pre-registered both-halves failure, and P1S0-CONTROL could not move it. **16 trials to sweep a space whose corners are all measured and whose vehicle is closed is a search, not a question.**

If exactly one cell is to be run, run 60–90 DTE $\times$ 0.85–0.95 delta **alone**, 1 trial, framed as DEEPTM-FIN's return-side companion — and pre-commit that its failure closes the tenor question permanently rather than prompting a fifth tenor. DEEPTM-FIN's own scope note applies verbatim: it is a **cost** measurement and says nothing about returns.

MB3 — Event ownership: real, mean-driven, and untradeable by this operator — and the deciding arithmetic costs zero trials

EV: MEDIUM-HIGH (it closes a live ambiguity). Trials: 0 for the decisive arithmetic; 1, options, only if a disclosure is wanted. Prior: ~85% it closes.

The record, corrected against the commission's own framing. The commission describes 017C4's effect as having nowhere to live. The ledger is sharper than that: 017C4 is DONE – REJECTED, and solely on a bar of my own that was broken — c1, c2 and c4 all pass and it fails on c3. And the register's own post-mortem (CLAUDE.md, session 62) diagnosed two of its three bars as not measuring what they were written to measure: B1's trades-per-year axis is **unpassable by construction** for a subset of the book it is drawn from, and B1/B2's name axes re-measure the **29-name foreign-issuer coverage hole**. **Only B3 measured its property, and B3 passes.**

So the effect stands and the rejection does not rest on it. Measured: a call spanning the next announcement earns **+10.30% against +5.50%** on 27,350 random-entry control trades, **+4.79pp**, positive in both halves, sign test **z +2.054, p 0.040**.

And the same register measured why it cannot be traded here. It is a **MEAN** effect, not a median one: DTE-matched, **median-vs-median is +0.40pp** (–51.41% against –51.81%). The typical trade is a near-total loss either way. Harvesting a mean carried by a thin right tail requires many small independent positions held without intervention — which is exactly the configuration 011 measured to end below its starting equity at a \$50k account and cap 10.

The decisive test that has never been run, and it charges nothing. 012's Kelly and ruin machinery is built and its inputs are the banked per-trade returns. Ask it one arithmetic question: **at what account equity does the cap-10 ruin arithmetic permit an earnings-spanning book to end above where it started?** That is a computation on banked distributions with no hypothesis and no bar — the S25 / X7RECON class, **zero trials**.

Kill condition. If the required equity exceeds **\$250,000**, the answer is final for this operator and the family closes permanently, with the effect recorded as real and out of reach. Below it, the question becomes a live design and needs its own blind register.

My argued verdict, either way as the commission asks. *Real, mean-driven, untradeable at this account size.* The honest home for it is a **disclosure in MA28's class** — "options spanning an earnings announcement have historically had a higher mean return and an essentially unchanged median; the mean is carried by a small tail" — not a strategy family. 011 governs and nothing here licenses trading it.

MB4 — Expression and financing: the door is closed, and the closing argument is arithmetic

EV: MEDIUM (a closed door, argued from measurements, is a legitimate deliverable). Trials: 0.

The commission asks whether ANY expression question survives. Taken one at a time:

expression argument	status	the measurement that closes it
Cheaper leverage than margin	CLOSED at today's brokers	DEEPITM-FIN: all-in 701.87 bps/yr (financing 342.35 + roll 340.06 + commission 3.57) at median DTE 73 and 5.0 rolls/yr. Cheaper than exactly one of three retail cards — Robinhood standard at rf + 995 — and more expensive than the two an operator would use (Robinhood Gold rf + 420, IBKR Pro rf + 150).
Financing improves with tenor	CLOSED as unownable	The owned cache is hard-capped at 200 DTE (frontier \$2a); the Tier E pull reaches 858 DTE but only for 2016–2018 . And the frontier's own reading is that 60–90 DTE, the shortest tenor, is the worst case — which is what makes the measured number the relevant one.
Defined-risk sizing	CLOSED by mechanism	O11's ruin is driven by opportunity refused at the concurrency cap , not by tail loss. Capping the tail with a spread does not add capacity in the crowded weeks where the expectancy lives; it reduces it, because a spread ties up the same slot for less premium.
Tax	ALREADY ANSWERED	U5 is DONE.
Collateral	NOT A CONSTRAINT	The operator holds no position that a share purchase would breach; there is no margin requirement for an option to relieve.
Alpha in the expression	CLOSED	P1S0 failed at its power anchor in both halves at all three horizons and closed the options-expression family; P1S0 - CONTROL returned NULL and could not move it.

Verdict: expression is CLOSED. The one re-open condition is a *financing* fact and not a research one, so it needs no trials and it can be stated as a number: **if the operator's borrowing rate ever exceeds rf + 702 bps, the deep-ITM call becomes the cheaper leverage** and DEEPITM-FIN has already priced it. Today's Robinhood Gold and IBKR Pro cards are both below that line. Nobody should spend a trial re-deriving this.

MB5 — The record correction the re-open list asked for landed on two rows of three

EV: LOW-MEDIUM (free, and it is a record integrity item). Trials: 0.

The blind re-open list found that three registers priced non-entry-day chains from the **mutable** data/options store without disclosing it, and recommended the disclosure be corrected *"regardless of whether anything is re-run"*. Measured on today's ledger: **06 and 07 now carry it; U1 does not**. U1's row still describes its grid — *"182 names x 39 rebalance dates ... 6,811 cells to 5,186 trades"* — without saying which store served it, while scripts/u1_entry.py's own docstring says *"READ-ONLY on data/options/"* and its dates are rebalance dates the freeze never covered.

Action: one sentence on U1's note. **No re-run** — the re-open list's reasoning stands (U1 failed all four pre-registered conditions and every decile's median trade sits between –52.5% and –54.3%, which no chain source moves).

MB6 — Tier C's small caps are an options-side object too, and the flag is not optional

EV: MEDIUM. Trials: see MB17 (mandate 4 owns the equity half). Pointer item.

The 420 never-tried optionable names now hold **982 units with data, 50.6 M rows, 2016–2018** — a population no options study has ever touched, and the only one that is small-cap. Two hard constraints must be written into any register before it runs, both from the census: **28 of 982 units (2.9%) across 14 symbols carry another company's option data** (pre_panel_history, with panel_first_year on every row — FOXA, IR, VG, CR, AZPN and nine others), and treating them as the modern company is a live way to get a wrong answer; and **278 units are empty_vendor**, so a name's absence is a fact about the vendor rather than about the market. The equity-side question this population uniquely answers is in MB17.

MANDATE 2 — COMBINATIONS AND AMPLIFIERS

MB7 — The incremental-IC gate is defective on this panel, and it gates everything else in this mandate

EV: VERY HIGH (free, and six other items depend on it). Trials: 0 — a specification correction.

The commission requires every proposed cross to pass "the PEAD/U2 template": residualise the candidate on the seven weighted incumbents, then test the residual IC. **On this panel that template silently deletes 20 of 69 rebalance dates.**

Measured directly from `panel_corrected_69d.pkl`, reproducing MA58-SEAS independently:

basis	rows kept	dates with ≥ 20 names	first such date
all seven incumbents (complete case)	66,444 (58.31%)	49 of 69	2014-01-17
six incumbents, institutional dropped	92,540 (81.21%)	69 of 69	2009-01-15

The cause is one column: `institutional` has coverage **0.7172** and its first date carrying 20 or more names is **2014-01-17**. Every other weighted theme starts 2009-01-15.

Consequences, and they are not small. (1) Any incremental-IC register on this panel is a **post-2014 test** unless it says otherwise. (2) Its "early half" is not the panel's early half — of the 49 surviving dates only **28** precede 2021, and once a candidate's own coverage filter is applied on top (as MA58-SEAS found) that cell can fall below S18's 16-date floor, which is a second and independent reason such a design cannot return an interpretable both-halves verdict. (3) U2, MA31 and MA32 all used this template, so their reported early halves inherit it.

The fix, and it is a choice the register must make explicitly rather than inherit: either **(a)** residualise on the **six** full-window themes and report `institutional` incrementally as a declared secondary arm, or **(b)** keep all seven and state in the register that the test is post-2014 with a 28-date pre-2021 cell. Option (a) is better on power; option (b) is better on comparability with U2/MA31/MA32. **Pick one before running anything in this mandate.**

MB8 — MA28's crash flags as a position-SIZING haircut, judged on crash count and never on alpha

EV: HIGH. Trials: 1, equity (N 234 $\rightarrow$ 235; hurdle 3.3031 $\rightarrow$ 3.3042). Prior: ~50%, and the uncertain leg is the one that decides.

Mechanism, and it is why sizing rather than selection. MA28-CARD measured a **crash-rate ratio**, not an alpha: names tripping two or more of Beneish $M > -1.78$, Altman $Z < 1.81$ and top-decile external financing lost more than half their value over the next quarter at **2.6597% against 0.8743%, ratio 3.0422 $\times$** , replicating at 3.42 $\times$ early and 2.93 $\times$ late, with every window's observed value beyond the **maximum** of its own 500-draw permutation. And the effect **strengthens monotonically with size**, 2.010 $\times$ in the smallest quintile to **5.169 $\times$ in megacaps**, because the kept rate falls 14.5 $\times$ across quintiles while the flagged rate falls only 5.6 $\times$. A quantity of that shape maps onto **how much to hold**, not onto **what to hold**.

The statistic must be a name-level crash count, and the register must say so before it runs. S10-ACCT ran the exclusion version and failed on the **portfolio-drawdown** leg — and S10 had already measured why that leg can never pass: this book's worst peak-to-trough spans **exactly one 63-day period on every arm, COVID 2020Q1 at trough index 44 of 69**, which no name-level rule can move. So the primary is **the count of top-decile holdings suffering a > 50% quarterly fall**, weighted by the haircut, which is the object the flag demonstrably predicts.

Why this is not S10-ACCT in costume. Different intervention (a 0.5 $\times$ weight, not a deletion), different primary statistic (name-level crash count, not portfolio max drawdown), and the alpha leg is **non-inferiority** rather than improvement. The register must state all three.

Register. Haircut fixed at 0.5 $\times$ before the run and not swept. Primary: flagged-name crash count in the book, both halves. Secondary and gating: book top-decile alpha as a **non-inferiority** leg against x7's calibrated **1.8629pp** margin (MA19's recalibrated figure at $N = 224$ — and see MB31, this floor is one of the two that has already moved once and is due a re-derivation at $N = 234$). C7's eligibility sensitivity travels: **22.01% of rows carry fewer than two computable inputs** and sit in the base-rate group by construction, so the arm must be re-read on eligible rows only

as a declared sensitivity.

Kill condition. If the crash-count reduction is under **20% in either half**, the sizing family closes permanently. If alpha non-inferiority fails at the calibrated margin, the arm is REJECTED regardless of the crash result — a risk control that costs alpha is a trade, not a free lunch, and this register may not make that trade.

MB9 — MA28's flags as a short-put veto: REFUSED as stated, and the moneyness version is a new hypothesis

EV: LOW as stated. Trials: 2, options, only for the re-specified version. Prior: ~12%.

The commission floats the crash flags as a veto on the short-put side. **It is V6-OPT in costume and must be refused.** V6-OPT's decisive mechanism: *selling a 25-delta put is selling a 25% assignment probability by construction*, so the strike has already spent the risk edge — measured, assignment came back **25.30% healthy against 25.73% unhealthy**, a 0.43pp gap, while the unhealthy name paid **2.978% of strike against 2.550%** purely because its implied vol was higher. A crash flag is a risk signal; a delta-targeted put is blind to a priced risk difference by construction.

The one version that is not in costume is the one V6-OPT itself named as the obvious re-opening: **target MONEYNESSESS, not delta**, so the strike does not move with the name's own volatility. That is a **new hypothesis needing its own blind register**, and it must additionally argue past two standing closures: the options-expression family (P1S0) and the short-vol question (09). **I do not propose it**; I record that it is the only live form and that anyone proposing it inherits three unfavourable priors.

MB10 — V6-B's small-cap survival gradient: the honest use is a disclosure, not a surface and not a book change

EV: MEDIUM. Trials: 0 for the disclosure; 1, equity, for a sizing version I recommend against.

v6-B measured a large, replicated effect — healthy 20% drawdowns fall a further 20% **10.228pp** less often at HAC t -10.5847 , both halves, four to five times its own MDE in every window — and attached its own standing caveat: **-14.287pp in the smallest quintile against -3.787pp in megacaps**, so *"the claim is strongest exactly where the product is not"*.

A small-cap surface is the wrong answer, for two reasons already in the record. v6 measured that a drawdown on this panel is substantially an **inverse-momentum sort** (Spearman $+0.6642$ against the momentum theme), so a dip surface systematically surfaces names the live composite is marking down — two screens that disagree by construction. And the megacap quintile, the one the live hot list actually occupies, is the single quintile that **fails** v6-B's own both-halves leg.

The honest use is the MA28-CARD-UI pattern applied to a caveat instead of a claim: on the existing dip surface, state the gradient and where the displayed name sits in it. Zero trials, zero hypotheses, and it makes a limitation legible instead of leaving it in a handoff.

If a sizing version is wanted it costs 1 equity trial, and its kill condition is nearly pre-decided: v6-B already measured the megacap separation at **3.787pp**, so a register requiring $> 3.0pp$ in the megacap quintile is asking a question whose answer is on the record. **That is a reason not to charge the trial**, and I record it rather than proposing it.

MB11 — The optionable partition as a reported diagnostic, which is all P1S0 left it as

EV: MEDIUM. Trials: 0.

P1S0-CONTROL measured that the optionable subset is **worse early and better late**: the full 2,531-name panel beats the optionable subset by **+1.467pp (H=63)**, **+6.124pp (H=252)**, **+1.527pp (H=504)** over 2016–2020, and loses to it by **-9.704pp**, **-10.927pp** and **-1.555pp** late. The live hot list is largely optionable names, so this is a description of where the product's own universe has sat.

It may be reported and it may not be used. P1S0's gate closed the family on it, and the register's own defect note says the two effects **interact and its gate cannot separate them**. So the copy must say the partition is a description of the past and not a rule, and it may carry no forecast. Zero trials, and it belongs with MB39's disclosure-card family.

MB12 — MA58 landed, and the pattern across three nulls is worth more than any of them

EV: MEDIUM (a record item that constrains future registers). Trials: 0.

The commission asks me to check the ledger for MA58. It landed on 2026-08-18: MA58-SEAS — DONE — UNINTERPRETABLE, zero adopted, equity N 232 → 234. The seasonality cross is closed for the reason MA58 gives, and no further trial should be spent on it.

The pattern it completes is the finding. This panel has now produced **three** signals carrying genuinely new information that predict nothing measurable with it:

item	orthogonality measured	verdict
U2 — options-surface features	incumbents explain only 5.5%–8.8% of their variance (against 41.3% for gp_on_capital, 78.4% for ret_6_1)	all four arms REJECTED
MA31 / MA32 — parity deviation, open/close share	R ² on the seven incumbents 0.0438 / 0.0273 / 0.0361	all three NULL
MA58 - SEAS — annual-lag seasonality	mean R ² 0.0755 ; largest theme correlation −0.1511	UNINTERPRETABLE, and REJECTED on its own bars

So on this panel orthogonality is not evidence of value, and it has now failed as a motivation three times. Any future register citing "structurally orthogonal to everything in the panel" as its reason to expect an effect must cite these three against itself. That sentence is free and it will save trials.

MANDATE 3 — THE 2016–2020 PROBLEM

MB13 — The regime contrast is NOT ANSWERABLE on this panel, and here is the arithmetic

EV: VERY HIGH (it forecloses a whole family, rigorously). Trials: 0.

The commission asks whether there is a pre-registerable design that answers "*when does this composite work*" without fitting the regime to the answer, and instructs me to say so plainly if the answer is "not without another decade". **It is worse than a decade, and TIDEMARK's power-gate method is the instrument that says it properly.**

The inputs, all measured, none assumed. From the shipped artifact: quarterly top-decile alpha has mean **+1.794pp**, sd **3.298pp**, lag-1 autocorrelation **+0.0812** — so the design effect is **1.177** and n_eff/n is 0.850. From the panel's own date list: the documented quant-crisis window holds **9** rebalance dates and the P1S0 dead half holds **20** of 69.

The minimum detectable regime contrast, in annualised top-decile alpha:

contrast	n_A	n_B	SE	MDE at abs t = 2	MDE at the honest hurdle 3.3031
documented crisis vs rest	9	60	1.279pp/qtr	10.23pp/yr	16.90pp/yr
P1S0 dead half vs rest	20	49	0.949pp/qtr	7.59pp/yr	12.54pp/yr
plain halves	34	35	0.861pp/qtr	6.89pp/yr	11.38pp/yr

The published total top-decile alpha is 7.17pp/yr. Every cell in the right-hand column asks the regime *difference* to exceed the entire strategy, by a factor of between 1.6 and 2.4.

Read the other way round, which is TIDEMARK's more useful form — how much history would be needed at 80% power and the honest critical value:

regime alpha gap	quarters required per window	years each side
2.00 pp/yr	1,757.7	439.4
4.00 pp/yr	439.4	109.9
7.17 pp/yr (the whole published alpha)	136.8	34.2
10.00 pp/yr	70.3	17.6

regime alpha gap	quarters required per window	years each side
15.00 pp/yr	31.2	7.8

The panel supplies 17.3 years in total. The crisis window supplies 2.3.

Ruling, in TIDEMARK's grammar: NOT PERMITTED. No regime-conditioning design on this panel can return an interpretable verdict, and this does not depend on the multiplicity correction — even at an uncorrected $t = 2$ the crisis contrast needs a 10.23pp/yr gap. **The naive fix the record warns about most is not merely dangerous here; it is arithmetically futile**, which is a stronger and more durable reason to refuse it than "it is p-hacking".

What this forecloses: every regime-conditioning register on this panel, permanently, until forward time accrues at one quarter per quarter. **What it does not foreclose:** MB14.

MB14 — The one pre-registerable thing left: a three-state diagnostic on an externally-dated window

EV: HIGH (1 trial converts the project's largest open wound into a dated, powered refusal). Trials: 1, equity (N 234 → 235; hurdle 3.3031 → 3.3042 — 0.0011 of a t). Prior: ~70% CANNOT-TELL, ~20% FAILS-THROUGHOUT, ~10% WORKS-OUTSIDE. I recommend running it.

The regime definition comes from outside the panel and from before this project looked, which is what makes committing to it blind meaningful. The 2018–2020 quant crisis is a documented, independently dated episode: Blitz, *The Quant Crisis of 2018–2020: Cornered by Big Growth* (Robeco, published February 2021) finds that essentially the only way to outperform in that window was to hold the largest and most expensive growth stocks, and that other factors worked only through implicit exposure to them — and that this episode was distinct from prior value drawdowns, which are better described as momentum rallies with collateral damage. **Replication status:** the episode itself is not contested; it is documented across practitioner research (Robeco, Man Group, First Sentier) and its dating is external to this project. What IS contested is any causal mechanism, and the register must claim none.

The discriminating fact, and it is why this design has content. P1S0-CONTROL's dead window is **wider** than the documented crisis. Measured on the panel's own dates: 2016-01-20 to 2018-04-20 holds **10** rebalance dates that are **outside** the crisis, and 2018-07-20 to 2020-07-22 holds **9** inside it. So the two hypotheses make different predictions on a real, disjoint sub-sample.

Design. Commit the window blind, before any scoring: crisis = 2018-06-01 to 2020-08-31, taken verbatim from the published dating and not tuned. Score the composite's **sorting** statistic — monotonicity and long-short, never a level, because P1S0-CONTROL's own defect note is that its leg 2 "asked a **LEVEL** question when the item is about **SORTING**" and that is exactly why it returned NULL. Three states, fixed in advance:

• **WORKS-OUTSIDE** — the composite sorts over 2016-01 to 2018-05 and fails only inside the crisis.

• **FAILS-THROUGHOUT** — 2016–2018 fails too; the crisis explanation is refuted and the failure is longer and unexplained.

• **CANNOT-TELL** — the pre-committed power rule fires. MB13 says it very probably will, and the register must carry MB13's table so the reader can see that this was known before the run.

Kill condition. The register pre-commits that a **CANNOT-TELL closes regime work on this panel permanently**, and that the only admissible re-open is accrued forward time — not a new window, not a finer statistic, not a different regime variable. Without that clause this item becomes the first of five.

Why spend a trial on a probable CANNOT-TELL. Because the alternative is leaving the project's single most consequential open finding in a state where any future session can re-litigate it cheaply. A dated, pre-committed, powered refusal costs 0.0011 of a t on the hurdle and removes a recurring cost permanently.

What the product may honestly DO with any verdict: almost nothing, and the constraint is already shipped. v3 measured that **where an individual name sits inside the top decile is not distinguishable from chance** (holding on 45 of 69 dates) and that the **group-level** result holds on only **21 of 69** — "which is why it may never be stated as a standing property". So a regime verdict may not modify a name's score, may not gate the hot list, and may not become a standing claim. The only honest action is a **dated disclosure carrying the power statement**, in MA28-CARD-UI's pattern: the composite did not sort over this window; the window overlaps a documented industry-wide factor drawdown; we cannot tell which, and here is the arithmetic (34 years per side) that says why.

MANDATE 4 — UNTOUCHED INSTRUMENTS

MB15 — The tick cache's `exchange` field has never been read by any study, and it is the retail axis

EV: HIGH (a genuinely unread field on an owned 70 M-print instrument). Trials: 2, options (N 300 → 302). Prior: ~15%.

Measured, read-only, on three sampled payloads (NVDA 2016-05-13, NVDA 2016-09-21, AAPL 2016-12-15). Every print carries `exchange` with **14 distinct venues**, plus `bid_exchange` and `ask_exchange` (14 and 13–14 distinct). 014's five features were `signed_volume`, `pc_flow_imbalance`, `block_share`, `unusual_volume` and `sweep_share` — **none touches venue**. 010/018 used condition codes (18, 35, 106, 0, 95 dominate) and `size`. **No study in the corpus has read `exchange`.**

Why it matters. Bryzgalova, Pavlova & Sikorskaya, *Retail Trading in Options and the Rise of the Big Three Wholesalers* (Journal of Finance 78(6), 2023) identify retail options flow through wholesaler routing and find retail is **> 60% of total options volume**, prefers cheap weekly options at an average **12.6% bid-ask spread**, and **loses money on average**. Bogousslavsky & Muravyev's *An Anatomy of Retail Option Trading* extends it. **Replication status: published in the JF, widely cited, and the loses-money direction is robust** — but the identification is a wholesaler flag this vendor does not supply, so a venue-based classifier here is a **PROXY and must be labelled one on every output**.

Corroborating shape already visible in the cache: `size == 1` is **24.4%–31.6% of prints** and the median print is 3–5 contracts, consistent with a retail-dominated tape.

The conditioning caveat, named per the commission's instruction. The cache is **alert-days only** — 186 symbols, 1,574 dates, and the alert screens on unusual volume versus open interest and low IV. Retail share is mechanically higher on high-volume days, so this sample is **selected toward the retail-heavy tail**. Two consequences the register must carry: a null may be **range restriction** rather than absence, and a positive is measured on the part of the distribution the alert already selects and **does not generalise to ordinary days**. 010 measured the harder version of the same limit: the next session is cached for **0 of 3,870 trades** and the exit day for **0 of 3,870**.

Kill condition, and it fires before any outcome is read. The venue → retail mapping must be validated against an external benchmark first: it must reproduce the published retail share on the pooled cache to within ±15pp of BPS's ~60%. **If it cannot, the arm is VOID before scoring.** That is the strongest available kill, because it tests the instrument rather than the hypothesis.

Register. Fix the mapping and the validation before any outcome; both halves; the range-restriction control (alert-day retail-share distribution against 010's control book) as a gating control run and read in a **separate pass** — 010's own process defect, not repeated.

MB16 — VPIN is CONTESTED, and the dispute names which version to build

EV: MEDIUM. Trials: 1, options. Prior: ~10%.

Easley, López de Prado & O'Hara's VPIN is proposed as an order-flow-toxicity measure. Andersen & Bondarenko's critique is specific and it is the useful part: against a benchmark using quotes and trades, the **Bulk Volume Classification** scheme is **inferior to a standard tick rule**, and VPIN predicts volatility largely because rising volatility induces systematic classification errors in BVC. The authors dispute the characterisation. **Replication status: CONTESTED, and the contested component is the classifier.**

That is exactly the component this project already does properly. 014 built a quote-based Lee–Ready classifier and measured it classifying a median **98.54%** of eligible prints — and found and fixed a real defect in it (the tick test needs the previous *different* price). So the only version worth building here is **quote-classified VPIN**, which the critique does not reach.

Kill condition, before scoring. If quote-classified VPIN correlates above **0.90** within date with 014's already-null `signed_volume` or `unusual_volume`, it is those features renamed and the arm is **withdrawn before any outcome is read** — the `u2 skew_25d` precedent, where a separately-named arm turned out to be an existing column's negation and was killed before the register rather than after the verdict.

MB17 — Tier C's 420 small caps: the one question uniquely answerable there

EV: MEDIUM-HIGH. Trials: 1, equity. Prior: ~25% it returns an interpretable verdict at all.

What is unique. V6-B measured its survival effect **strongest in the smallest quintile** (~14.287pp) and weakest in megacaps (~3.787pp). P1S0 measured the optionable subset **worse early** — over exactly the 2016–2020 window. Tier C supplies **982 units of 2016–2018 chain data for 420 optionable names the breadth miner never covered**, which are small by construction (2.6 MB per symbol-year against a megacap's 8.8 MB). That is the **only** population where the composite's dead window, the small-cap end of V6-B's gradient, and options existence intersect.

What is answerable and what is not. Answerable: *does the composite sort the small optionable names over 2016–2018, where the record says it should be strongest and where it also says the optionable subset is weakest?* Not answerable: anything requiring 2019+ (Tier C is 2016–2018 only), and anything requiring a name to be in the panel before its debut.

Two hard requirements from the census, both mandatory before any number. Filter `pre_panel_history` — **28 of 982 units across 14 symbols** carry another company's option data with `panel_first_year` stamped on every row — or resolve those symbols against a point-in-time identifier. And treat `empty_vendor` (278 units) as a vendor fact, never as a market fact.

Kill condition. MB13's arithmetic applies with less data: if the covered small-cap cross-section has fewer than **16 dates per half** (\$18's floor) or its power control — the panel's own known-real signals on the arm's actual rows, per MA31/MA32's three-population method — fails a $t = 2.0$ bar, the result is **UNDERPOWERED, not null**, and the population is closed to further equity registers.

MB18 — The S23 valuation panel's genuinely unread object: the implied-growth expectations gap

EV: HIGH — the cheapest genuinely-new equity hypothesis in this audit. Trials: 1, equity. Prior: ~15%.

Measured, read-only, on `panel_s23_fairvalue.pkl`: 108,241 rows, **69 dates, 2,441 names, 2009-01-15 → 2026-01-28**, with `implied_growth` at **100.00%** coverage and `base_growth` at **100.00%**. Both are solved from price and filed fundamentals **as of the date** — point-in-time. The gap `implied_growth - base_growth` is *how much more growth the price demands than the company's own recent trajectory*: a reverse-DCF expectations measure, and nothing in this project has ever scored it.

Mechanism. Expectations errors rather than risk: prices embed a growth path, the path is systematically too optimistic where it is most extrapolated, and the error is what earns. The project's own measurement of the level agrees — `implied_growth` has median **0.164** against `realized_growth`'s median **0.061** on the 81,633 rows carrying both — and `calibration.implied_growth_realization` already exists to report it.

The look-ahead trap, named before anything is built. `realized_growth` is **forward** three-year growth. It may **never** enter a signal; it is an ex-post attribution only. Any arm that touches it is void by construction, and the register must pin that at source — the MA31 precedent, where `dividends.spot_from_parity` fed back as the spot would have made the arm identically zero and nothing would have raised.

The costume risk, named with its own kill. A reverse-DCF implied growth is monotone in price over fundamentals, so it may be the value theme renamed. **Mandatory C6-style control: mean per-date Spearman against the value theme, and the arm is WITHDRAWN before any outcome is read at $abs\ \rho > 0.60$.**

Register. The incremental-IC gate **as re-specified by MB7** (this is one of the six items that gate depends on); both halves; the correlation control above run and read in a separate pass; `implied_bounded` (a 3-state solver flag at 100% coverage) reported as a data-quality partition rather than silently pooled.

Inherited caveats. The panel's `fair_value` is S23's reconstruction and **not** what the live site published on the day (MA26-C); and S23 itself found and fixed the path fetching **live** Yahoo prices to value 1999, so any re-build must assert zero network calls as S23 now does.

MB19 — MA55's lens-disagreement width is buildable, and the `w_floor` is the whole design

EV: MEDIUM. Trials: 1, equity. Prior: ~12%.

Verified read-only: all three valuation lenses are present on **80,689 of 108,241 rows (74.55%)** — `dcf_ps` 99.87%, `comps_fv` 100.00%, `growth_ps` 74.55% — so MA55's premise holds exactly as its design record states. Its measured

width distribution is p05 **0.1195**, median **0.8777**, p95 **4.1069** — with a **maximum of 3,585**, four orders of magnitude above the median.

That maximum is the design. Without a floor the arm stops being a precision-weighted mispricing signal and becomes a **disagreement screen**, which is a different hypothesis. The `w_floor` is a free parameter and the register must pre-commit it **and** prove the verdict insensitive across a pre-declared $\pm 50\%$ band.

Kill condition. If the verdict moves anywhere inside that band, the arm is **VOID** — a result that depends on a free parameter is MA58-SEAS'S C-DEPTH finding in a new place, where $K = 5$ and $K = 10$ told opposite stories and only the pre-committed one carried a verdict.

MB20 — MA57's one-line change now has a consumer, and it is the one construction `S3` never built

EV: MEDIUM-HIGH. Trials: **1**, equity, plus a two-column `_KEEP` change. Prior: **~20%** — the highest prior on any equity arm in this audit, and still under a quarter.

MA57 refuted the audit's data blocker (the export has **24 columns**, both `ownername` and `transactioncode` present, **5,636,964 rows**, zero missing on all **124,181 open-market purchase rows**) and then **correctly declined** the `_KEEP` change for having no consumer. The COVERAGE RULE's discipline is to add source columns when the signal that needs them is added.

Here is a consumer. Cohen–Malloy–Pomorski's routine-versus-opportunistic insider split needs exactly those two columns, and MA57 already computed the split on the export: **42,537 of 87,318** (`ownername`, `ticker`) pairs = **48.72% routine** on all coded rows. `S3` **rejected three insider rebuilds and none of them was this one** — `S3A` dropped the buys bonus, `S3B` scaled by market cap, `S3C` split into two z-scored inputs. The routine/opportunistic split is a different construction, and it is the one the published literature says carries the signal.

Kill condition, and it is a coverage rule not an outcome rule. `transactioncode` is **absent on 1,544,490 rows (27.40%)**, and a blank code can be classified neither way. Pre-commit that if the classifiable share of the panel's own insider rows falls below **60%**, the arm is **UNDERPOWERED**, not null, and the `_KEEP` change is reverted rather than kept for a dead consumer.

The unflattering prior, stated because it is the record's own. `S3` measured that the pure indicator "*has an insider score at all*" carries a **larger** absolute t (+1.4471) than the insider theme's own direction (−0.2259). Neither is significant; the comparison is the point, and it should be quoted in the register rather than discovered afterwards.

Also fix in passing (reported, not repaired): `_KEEP["insiders"]` requests a column named `date` that the export does not have — it is `transactiondate` — so `filingdate` or `date` has a fallback that **can never fire**. MA57 found it; it is still there.

MB41 — Intraday momentum is NOT TESTABLE HERE, named so nobody proposes it

EV: LOW (a disqualification, and disqualifications are cheap). Trials: **0**.

Gao, Han, Li & Zhou (*JFE* 2018) find the market's first half-hour return predicts its last half-hour return, stronger on volatile, high-volume, recession and macro-release days, and significant out of sample as well as in. It is a plausible thing to reach for once someone hears "70 million prints".

It cannot be tested here, and the reason is the instrument rather than the result. It is an **index/ETF** phenomenon measured on SPY half-hour returns. This project's tick cache is **single-name options prints on alert days**; the chain cache is single-name equity options only; and the frontier already established that **no index chain is owned** (P6, NEEDS-DATA). There is no equity tape here at all.

Marked **NOT-TESTABLE-HERE** so it does not consume a session's reading before someone reaches the same conclusion.

MANDATE 5 — CROSS-POLLINATION FROM TIDEMARK

MB21 — The Boudoukh–Richardson–Whitelaw null, and `S22`'s null is mis-specified in the direction that manufactures its result

EV: HIGHEST IN THE AUDIT. Trials: 0 for the diagnosis (done, below); 1, infra, for the persistence-preserving null. Prior: ~55% that at least one horizon crosses.

The instrument. BRW (RFS 2008): with a persistent regressor and overlapping long-horizon returns, R^2 and long-horizon β rise **mechanically under a no-predictability null**. TIDEMARK reproduced it on its own data and the result is stark — the actual data's R^2 ratio rises **more slowly than the median of pure noise at every horizon** (3, 5, 7, 10, 15 and 20 years), so what looks like long-horizon predictability in the famous CAPE chart is, on that test, entirely a mechanical artifact.

Valquo's exposure. S22 is the term-structure item: annualised top-decile alpha essentially flat from 3 months to 2 years, cumulative alpha reaching +10.20% at eight quarters, classified CONSTANT-RATE on $R(8) = 6.195$ against a 6.0 bar, corroborated by "*median rank IC rises with horizon, +0.034 $\rightarrow$ $\sim$ +0.072*". Its null is a per-horizon `fixed_weights_null` built from `placebo_panel`, whose method is stated in its own docstring: "*Within each rebalance date, permute the signal columns AS A BLOCK across the names present.*"

A within-date permutation destroys the signal's time-series persistence completely — and that is the one property BRW's artifact requires. Measured here, read-only, on `panel_corrected_69d.pkl` with the deployed flat 1/8 weights:

object	lag 1 (63d)	lag 2 (126d)	lag 4 (252d)	lag 8 (504d)
real composite, per-name rank autocorrelation across rebalances	0.5802	0.4859	0.4514	0.4099
placebo_panel, seed 1000 / 1001 / 1002	-0.0011 / +0.0010 / +0.0005	—	—	—

(68 date pairs, median 1,548 matched names per pair; real lag-1 median 0.5718, min 0.4073, max 0.7109.)

Per theme, lag 1: size **0.9915**, capital_discipline 0.8882, value 0.6981, quality 0.6786, momentum 0.6414, insider 0.3152, institutional 0.1181. **The most persistent input by a distance is size — and x3 measured that size carries the composite's entire statistical significance** while having the worst theme IC. The two facts compound in the same direction.

So S22's null is a null for a signal with no memory, applied to a signal whose memory is 0.41 at two years. It cannot generate the mechanical rise BRW describes, and the shortfall grows with horizon — which is precisely the axis S22's headline lives on. Its $H=63$ result is untouched (no overlap, lag 1, horizon equals rebalance); everything beyond it is scored against a null that is too easy by an unmeasured amount.

This is product-consequential, which is why it ranks first. S22-DISPLAY IS DONE — SHIPPED and puts the registered sentence, its caveats and the misuse warning on the hot-list card, the per-name attribution panel and /methodology, including the rank-IC-rises-with-horizon corroboration (0.0336 at one quarter to 0.0655 at two years) — the exact statistic BRW's null predicts under no predictability.

The fix. Build a **persistence-preserving null**: permute each name's whole signal *time series* (or block-permute dates) rather than permuting within dates, so the placebo retains the real signal's autocorrelation while destroying its association with forward returns. Validate it the way `placebo_panel` was validated — assert the placebo's per-name lag-k autocorrelation matches the real signal's within tolerance, and assert its association with `fwd_ret` is nil. Then re-run S22's per-horizon comparison against it.

Trial accounting, and it needs deciding explicitly rather than assumed. Constructing and validating the null is **infrastructure** — 1 infra trial on the HACFLOOR / X7RECON precedent, and infra N gates no published claim. Re-scoring S22 against it is a **re-measurement of a landed claim on a new instrument**, which the discipline explicitly permits (a re-open needs new data, a new instrument, or a new design — this is a new instrument). It is not a new search and should not be charged as one, but the register must say so before running rather than after.

Kill condition, pre-committed both ways. If the persistence-preserving null's $H=504$ alpha floor lies **below** S22's observed value, S22 stands, the record is strengthened, and the item closes. If it lies **above**, S22-DISPLAY's two-year copy must be **withdrawn or re-scoped**, and the register must commit to that withdrawal in writing before the number is read. Without that clause this becomes an invitation to reinterpret.

What it does not say. It does not say S22 is wrong. It says S22 has never been tested against the null that its own headline shape most requires, that the null it used cannot produce that shape, and that the gap between the two is measured at 0.58 falling to 0.41 rather than argued.

MB22 — Port TIDEMARK's effective-n / required-n gate, so every register states its MDE before running

EV: HIGH. Trials: 1, infra. Cost: ~1 session.

What TIDEMARK has that Valquo does not. Valquo has `statistics.effective_n(n, rho)` (an AR(1) closed form) and, in the options lane, R3's design effect scored against a **shuffled null** (deff 2.1837 against a null p95 of 1.1898 — and R3's own lesson, that a raw design effect near 1.8 arises from pure sampling error and must never be quoted without its null, is already learned here). What Valquo does **not** have is the last step: the conversion from available observations to a **required-n bar**, `required independent observations = ((sqrt(2 ln N) + z) / effect)^2`, measured per series against a simulated no-predictability null carrying that series' **own fitted persistence**.

Which Valquo claims change status under it. MB13 is the demonstration: the regime family goes from "open and tempting" to **NOT PERMITTED**, with the number 34.2 years attached. v6, s19 and v6-B each computed an MDE by hand in their own registers, inconsistently and after the fact; this makes it a standard field. MA58-SEAS's three-population power decomposition is the same instrument built ad hoc for one item.

Validation, borrowed intact. TIDEMARK's own gate reproduces its charter's power table exactly (IR 0.20 → 196.0 against a printed 196; IR 0.30 → 87.1 against 87), which is the positive control any port should reproduce first.

Kill condition. None — it is infrastructure and infra N gates no published claim. But it must ship with TIDEMARK's own recorded deviation attached: `POWER_GATE.md §5.1` records that the validation its pre-registration asked for **could not be run**, because the column it was to be checked against is `n/h` (non-overlapping window count) and not a design-effect quantity at all — two different quantities sitting in adjacent columns of one table. A port that re-creates that comparison re-creates the error.

MB23 — Port and verify the Hodrick (1992) 1B estimator as a cross-check, not a replacement

EV: MEDIUM. Trials: 1, infra. Cost: ~1 session.

TIDEMARK verified 1B against Wei–Wright's published Monte Carlo to fixed tolerances — **23 cells, max abs deviation 0.016, mean 0.007 at alpha = 0, the only case 1B is valid for** — after finding its own first implementation wrong (it summed regressors while keeping the h-period residual, which is not 1B, and returned $t \approx 0.3$ at every horizon against a bootstrap $p \approx 0.018$). Post-fix, the three instruments agree: Hodrick ≈ 2.5 , design-effect-adjusted OLS ≈ 2.3 , bootstrap consistent.

Which Valquo claims change status: honestly, probably none at H = 63 — there is no overlap there and the HAC lag is 1. The claims genuinely exposed are s22's long horizons, where the HAC lag runs to 7 and the overlap is severe. **So this is MB21's companion instrument rather than an independent item, and it should be run with it or not at all.**

Kill condition. If Hodrick and Newey–West agree within 10% on the shipped $H = 63$ statistics, the port is recorded as a validated cross-check and **no claim moves** — which is the expected and desirable outcome.

The correction that must travel with it. TIDEMARK's `POWER_GATE.md §5.2` records that its pre-registered cross-check criterion was **misspecified** — it asked for the 97.5th percentile of abs t to sit within 10% of 1.96, which is the wrong quantile, and the rule **failed on the verified case**. The corrected statistic is the rejection rate against its nominal 0.05. A port that copies the criterion instead of the correction ships a rule that flags known-good cases.

MB24 — Data flowing from TIDEMARK: OUT OF SCOPE, and here is the fusion register I am declining to propose

EV: N/A — a deliberate refusal. Trials if ever run: 1 in EACH counter.

The commission requires that any TIDEMARK series conditioning any Valquo quantity comes with a fusion register or is marked out of scope. **I mark it out of scope**, for a reason that is arithmetic rather than procedural.

Why. The only plausible fusion is a rates or credit regime series conditioning the equity composite. MB13 measures that Valquo has **no power for a regime contrast at all** — 34.2 years per side against 17.3 available. Importing a conditioner for a question that cannot be answered buys nothing and costs a trial in **both** books, permanently: TIDEMARK's `TRIALS.md §5` forbids revising `N_booked` downward, so the charge there is irreversible, and it would

raise the critical value in a project that has already ruled Phase 2 **NOT PERMITTED**.

What the register would have to contain, recorded so a future session does not have to reconstruct it. (1) The trial charged in both counters, and Valquo's unified domain — declared in `research_log.DOMAINS` and reading **zero** — is the correct home, which would make it the first row ever charged there and would force R4's permanent residual to be decided. (2) TIDEMARK's barred list inherited whole: `us_equity_caey` is **BURNED** as its trial #0 and demoted by `PREREG_ANCHOR_SELECTION.md` because its percentile is partly a clock. (3) The display-versus-feed line: FRED series are public domain and may both feed and display; **Case-Shiller (S&P copyright, explicitly not redistributable), NAREIT (terms unverified) and Ken French may not be displayed raw**, only as derived statistics.

MANDATE 6 — TIDEMARK AS A SURFACE

MB25 — The integration, designed: a linked page, derived statistics only, and one non-vacuous test

EV: HIGH (Don has asked for it; it is a product item with no research risk). **Trials: 0. Cost: ~1 session.**

Lane: app-fixer (`valuation/web/`). Not the edge lane — nothing here touches a verdict.

Form: a linked page at `/tidemark`, not a tab inside the hot-list surface. The two projects have different verdict grammars and different denominators; a tab implies they are one product, and a reader who averages a Valquo statistic with a TIDEMARK one has been misled by the layout.

Files.

file	role	precedent
<code>valuation/web/tidemark_surface.py</code> (new)	owns the copy and the derived statistics; nothing else may hold a TIDEMARK string	<code>score_confidence.py</code> , <code>hold_horizon.py</code> , <code>accounting_risk.py</code>
<code>valuation/web/templates/tidemark.html</code> (new)	render only	<code>methodology.html</code>
a route in <code>valuation/web/app.py</code>	serve it	the existing <code>/methodology</code> pattern
<code>tests/test_tidemark_surface.py</code> (new)	the pin, below	<code>tests/test_hold_horizon.py</code> (19 tests, fails if either side is reworded)

What crosses the boundary — and TIDEMARK's own design already makes this cheap. Only derived statistics: **percentile, episode count, effective n, band half-width, tier label, refusal reason, vintage, market and anchor names.** Verified read-only against the generated `dashboard/index.html`: its `bignum` elements are percentiles and $\pm$ band half-widths, its cards carry tier captions and refusals, and **no raw series level appears anywhere on the page.** The builder's own docstring states the rule — *"No forward-return numbers. None."* — and the two bands it does show are $SE(p) = \sqrt{p(1-p)/n_{eff}}$ and $1.96/\sqrt{episodes}$, neither of which requires a licensed value. **So the licence constraint is satisfied by TIDEMARK's construction, which is the cheapest possible compliance argument and should be cited as such.**

The captions ship VERBATIM, on the `v3/hold_horizon.py` precedent: every sentence appears word-for-word in the source document and a test fails if either side is reworded.

The explicit copy the commission requires, in TIDEMARK's own words rather than a paraphrase: *"NO PHASE-2 QUESTION MAY BE ASKED ON THIS DATA ... the best-placed market has 82 of the 155 independent years required, and the worst has 25."* Plus its more useful inverse — the edge would have to run at IR 0.41 to 0.74 for this data to detect it, against a 0.2–0.4 plausible range. **A dashboard shipped without that sentence reads as a signal.**

The one test that pins "no raw series reaches the payload", and why an allowlist alone is not enough. Name it `tests/test_tidemark_surface.py::test_no_licensed_level_reaches_the_payload`. It has two halves: an **allowlist** of emitted keys (`market`, `anchor`, `percentile`, `episodes`, `n_eff`, `band_sigma`, `tier`, `refusal`, `as_of`) asserting the payload contains nothing else; and a **positive control** — feed the builder a fixture whose series level is a distinctive sentinel (e.g. `123456.789`) and assert that string appears nowhere in the rendered HTML. The allowlist can pass vacuously if the builder emits nothing; the sentinel is what makes it a measurement. That distinction is this project's own most-repeated test defect (`v6`'s `C8` reading a column name the panel does not have; `MA5`'s guard firing

on its own documentation; MA23's stale-path guard blind to the import syntax the codebase writes).

Kill / staleness condition. TIDEMARK's artifact carries a vintage (2026-08-17). If it is older than a pre-committed number of days, the page renders the vintage and a staleness note **instead of** the number — the `index_track.gate_state()` pattern, where every unrecognised outcome fails toward "not passed" rather than toward a claim.

MB26 — The one thing that must NOT ship: a combined verdict, or an unlabelled pair of denominators

EV: MEDIUM (it prevents a specific, likely misreading). Trials: 0.

Valquo's register stands at **549 trials** with a critical value of **3.3031** (equity) / **3.3775** (options). TIDEMARK's stands at **66 trials** with **2.89**. A page showing both projects' outputs without naming both denominators invites a reader to compare or average statistics that were never measured against the same bar. **Both ns and both critical values go on the page**, and no statistic from one project may appear inside a sentence about the other.

MANDATE 7 — THE FACTORY, ROUND 2

MB27 — Board state is derivable, the derivation is already written down in prose, and nobody ran it

EV: HIGH. Build cost: ~half a session. False-alarm risk: designed to zero.

The evidence is a file in the repo root. `ma_in_flight.json` exists to answer "*which items are being worked RIGHT NOW*". Its own `_meta` carries: the purpose, the method ("*measured ... from local branches ahead of origin/main plus git worktree list*"), a `how_to_refresh` command, an honest caveat that uncommitted work is invisible to it, and a correction recording that the brief which commissioned it named five in-flight items and **none of the five had a commit anywhere**.

It is dated **2026-08-14**. It lists MA13, MA19, MA36, MA37, MA15, MA16, MA20, MA35. **Every one of them is now DONE**. The file is five days stale and 100% wrong, and it contains its own refresh instructions.

That is MA59/MA60's lesson one level up. Those items replaced `check_lanes.py`'s hand-typed import dictionary with a derived graph after measuring the literal at 13 keys / 40 edges against a real 118 / 546. The same disease is now in the board state: **a hand-typed snapshot of something git already knows**.

Proposal: `scripts/board_state.py`, **deriving all four ingredients with no hand-typed input.**

ingredient	derivation	value today
in-flight branches	<code>origin/worktree-*</code> not an ancestor of <code>origin/main</code> , plus local branches ahead	0
claimed items	ledger status cells matching <code>IN ?PROGRESS</code>	2 , and both are stale (B13 says so itself; D11's handoff says the miner is idle)
handoff freshness	each <code>HANDOFF_*.md</code> mtime against the newest commit touching it	reported
stale locks	<code>.git/index.lock</code> and worktree locks, with age	the six-day-stale lock incident is a fourth ingredient at zero extra cost

The cry-wolf rule, honoured explicitly. MA21 declined a proposed warning because it would fire on 41 legitimate rows and be switched off inside a week. **This script must not warn.** It emits a report with four counts and **exits 0 always**. The **only** failing assertion is one that cannot cry wolf: *the generated board file is older than the newest branch tip it claims to describe*. That is a staleness fact about a generated artifact, not a judgement about anyone's work — the same shape as MA13's committed-literal pin on `by_domain`, which is checkable, diffable and cannot produce a false positive.

Kill condition. If over four weeks the derived board and the hand-maintained one disagree on fewer than two items, delete the generated file and keep the hand one. A second copy of a fact that never drifts is worse than one copy.

MB28 — The staleness monitor already exists and has no clock

EV: MEDIUM-HIGH. Build cost: one scheduled-task registration — and it is Cowork's, not this lane's.

`scripts/checkout_drift.py` is built, documented and tested (`tests/test_checkout_drift.py`). Its own header records the measurement that motivated it: the shared checkout was **1 commit ahead and 514 behind** `origin/main`, and the one local commit — the dated PT-WRITER failure note answering an open ledger row — had been stranded since 2026-08-10. It deliberately reports and repairs nothing, because *"a guard that silently repairs is a guard whose failures are invisible."*

Measured: nothing invokes it on a schedule. Its only caller is `check_drift.bat`, a manual double-click. No workflow in `.github/workflows/` (`auto-scan.yml`, `land-agent-branch.yml`, `track-backup.yml`, `track-row.yml`) touches it.

So the alarm for "the relay dropped a packet" is itself gated on a human remembering. That is the mandate's thesis, again, in the guard built to answer it.

The fix cannot live in CI, and the reason is structural rather than a preference. It measures a **local** checkout, which no hosted runner can see; and MA11's auto-land policy **refuses any branch touching** `.github/`, so a workflow change needs a human anyway. The honest fix is a **Windows scheduled task on Don's machine** writing `--json` output to a fixed path the next session reads — **Cowork's lane** under the tool-routing rule, and it should be routed there explicitly rather than left as a wish.

False-alarm risk: low. It reports counts and exits non-zero only on a measured drift, which is a fact rather than a judgement.

MB29 — The prompt receipt, and this audit is its own example

EV: MEDIUM. Build cost: a convention plus ~5 lines inside MB27's script.

The failure named by the commission: the manager's roadmap listed a lane as in flight *on an intention rather than a pasted prompt*.

The cheapest mechanization with teeth is one the project already runs elsewhere. The register discipline requires a `PROMPT_*.md` committed **ALONE** and a strict ancestor of every measurement commit — precisely so that intent is provably prior to result. Apply the same shape one level up: **a lane's first commit on its branch is its prompt, committed alone, as** `PROMPT_<lane>.md`. MB27's script then reports, per in-flight branch, whether a `PROMPT_*` blob exists at its base.

Evidence it would have bitten, from this session. At session start `git status` read ??

`PROMPT_audit4_master.md` — **audit #4's own commission was untracked**, so at the moment this audit began, the instruction defining it was invisible on `origin/main` and no other lane could have discovered what was being worked. The seven existing `PROMPT_*.md` files in the root are the convention already half-observed; this makes it complete and checkable.

False-alarm risk: none. It is a reported count, not an assertion.

MB30 — What NOT to build, named so nobody proposes MA21's sibling

EV: N/A — a refusal. Trials: 0.

Do not build a warning on `src=auto` rows, on blank verdict cells, or on ledger rows lacking a handoff. MA21 already declined the blank-verdict warning with a measurement: **41 of 230 DONE rows carry a blank verdict and every one is legitimate**, because `build_ledger.py`'s own reading guide says blank means *"not measured, or measured and reported in different words — never 'we don't know'"*. A guard firing on 41 legitimate rows plus every prose verdict would be switched off inside a week. The substitute MA21 shipped instead — pinning the vocabulary literal against the documented list so two copies of one fact cannot drift — is the correct shape and is already in place.

MANDATE 8 — INSTRUMENT CALIBRATION AGE, ROUND 2

Deliver the staleness map; do not recalibrate.

MB31 — The staleness map

EV: HIGH (it is the map every other item quotes from). Trials: 0.

instrument	calibrated at	today's N	shipped value	honest value at today's N	status
HLZ hurdle, equity	N = 224	234	3.2898772171176964	3.3031261300040304	STALE — verdict unchanged (clears_hlz_hurdle: false); shortfall 0.6700 → 0.6832
HLZ hurdle, options	—	300	—	3.3775086897463940	quote this for any options claim
theme IC t floor	N = 84	234	2.7072	unchanged at 84, 129 and 224	insensitive so far — re-derivable from banked draws
long-short naive floor	N = 84	234	2.1437	unchanged at 84, 129 and 224	insensitive so far
long-short HAC floor	N = 84	234	2.2837	unchanged at 84, 129 and 224	insensitive so far
PBO p5 floor	N = 84	234	19.667%	unchanged at 84, 129 and 224	insensitive so far
top-decile alpha margin	N = 129	234	1.8629pp	DUE	moved once already (1.9532 → 1.8629 between 84 and 129)
top-decile alpha HAC floor	N = 224	234	2.0540	DUE	moved once already (2.2913 → 2.0540 between 129 and 224)
Deflated Sharpe floor	N = 224	234	0.6637	STALE BY CONSTRUCTION	sr0 is a function of N, so every draw moves at every N
Deflated Sharpe, the statistic	N = 224	234	0.7863	STALE BY CONSTRUCTION	same channel

Which recalibrations are DUE, and why exactly these two. MA19 established the mechanism: a p95 over 100 draws is set by the fifth and sixth largest values, so **whether a floor moves depends not on how many draws flip but on where they sat**. The two floors that have already moved once are the two whose flipped draws sat in the tail — the alpha HAC floor's flipped draw ranked **4th of 100**, which is exactly why it moved when the long-short floors did not. **Those two are the candidates for a third move and they are the two the shipped product's headline is judged against.**

The good news, and it makes the recalibration cheap. X7RECON banked the per-draw (margin, se) rows precisely so the adopt set at any N is arithmetic, and MA19 re-scored only **three** draws in ~400 seconds rather than re-running a 3.4-hour sweep. So the due recalibration is a bounded job, not a sweep — and RUN_RULES rule 9 is the reason it is bounded.

Which are provably insensitive: none. Four floors have not moved across a 2.7× change in N, and MA19 explained why, but *"has not moved"* is not *"cannot move"* — session 12 recorded that their survival was *"luck, not design"*, and on the alpha HAC floor the luck ran out. **Report them as insensitive-so-far and never as invariant.**

MB32 — Which CLAUDE.md comparisons now quote a bar at the wrong N

EV: MEDIUM (free; it is a documentation correction). Trials: 0.

The X7 CALIBRATED THRESHOLDS table's authoritative column is headed **N = 224 (QUOTE THIS)** and is now one denominator behind at N = 234. Nothing in it changes a verdict, but two specific sentence shapes are at risk and both are the defect MA19 already found once:

1. **Any sentence pairing a numerator at one N with a floor at another.** MA19's example is on the record — *"0.8674 vs the 0.7216 floor"* pairs an N = 116 Deflated Sharpe with an N = 84 floor. The Deflated Sharpe moves at **every** N, so this shape recurs automatically unless the text says which N both sides are at.
2. **The two DUE floors above** (1.8629pp and 2.0540), which are quoted in S10's, V2G's and MB8's reasoning. They are correct at N = 224 and should carry that label until re-derived.

Action: label the column N = 224 rather than "QUOTE THIS", and add one line pointing at the live count. Zero trials.

MB33 — The artifact–log drift is live at 10 / 8 / 1, and that is the system working

EV: LOW-MEDIUM (a "do not act" finding, which is worth stating). Trials: 0.

BACKTEST_RESULTS.json was generated 2026-08-14T02:23:56 and carries by_domain {equity: 224, options: 292, infra: 14} against a live {234, 300, 15} — a drift of 10 / 8 / 1, total 530 against 549.

This is correct behaviour, not a defect, and MA21 enforced it in the only direction that cannot cry wolf: trials accumulate, so the artifact may lag the log and may never lead it. It lags.

Do not re-run for this reason alone. A re-run costs 20–40 minutes, and this project's own memory records that a run overwrites the tracked repo-root artifact — so a re-run from a dirty or non-canonical tree is a live way to make the canonical file worse. Re-run when a claim's relationship to its bar would change; here, none does (clears_hlz_hurdle is false at both 224 and 234).

MANDATE 9 — SIMPLIFICATION, ROUND 2, AND THE PIONEER PAGE

MB34 — 64 merged worktree-* branches are deletable, with a one-line safety proof

EV: MEDIUM (pure hygiene, zero risk). Trials: 0.

Measured: git branch -r --merged origin/main | grep -c worktree- = 64; --no-merged returns only the two rescue/* refs. Every one of the 64 is an ancestor of origin/main, so deleting the ref removes no reachable commit and no history. Total remote refs: 68.

Safety argument: ancestry is the proof, and it is checkable in one command before each delete. This is the one deletion in the audit with literally no downside.

MB35 — The two rescue/* branches: tag first, then delete — and the reason is the record's own

EV: MEDIUM. Trials: 0.

ref	commits ahead	content
origin/rescue/main-41d7b12	1	41d7b12 — "PT-WRITER 2026-08-10: cannot write row — mechanism for daily prices not documented in repo"
origin/rescue/wip-main-c4a3939	2	the same commit plus d39ec84, a working-tree snapshot touching only HANDOFF_STATUS.md and LAZY_PRICES_COVERAGE.md

PT-WRITER closed DONE on 2026-08-18. So both refs are relics.

The safety argument is the interesting part. 41d7b12 is the dated failure note that CLAUDE.md cites as the decisive evidence for what blocked PT-WRITER — "the correct behaviour and the answer to the row" — and it is the only object carrying that message. Deleting the ref makes the commit unreachable and eventually collectable. So: create an annotated tag (archive/pt-writer-refusal-2026-08-10) pointing at it, verify the tag, then delete the branch. A tag is a permanent ref at essentially zero cost, and this project's own memory records the near miss of deleting something "because history preserves it" without checking that history does.

d39ec84 is a snapshot of two generated/handoff files and needs no tag.

MB36 — Documentation: 258 root entries, and the move is a rename with one real hazard

EV: MEDIUM. Trials: 0. Cost: one commit — but read the hazard first.

Measured at the repo root: **68 PREREG_*.md, 48 HANDOFF_*.md, 12 VALQUO_*.md, 7 PROMPT_*.md, 2 DESIGN_*.md**, 258 entries in total, 168 markdown files. Repo size 28 MB.

Do not delete any of them. The `PREREG_*` files are the register's evidence and the whole method depends on them being committed, dated and readable — and the repo is **public** as of 2026-08-16, so they are now the credential rather than the paperwork. The `HANDOFF_*` files are `RUN_RULES` rule 2's deliverable. The `PROMPT_*` files are MB29's convention already half-observed.

What is proposable: move them, as git renames, into `register/` and `handoff/`. Precedent MA23, which moved twelve modules as renames with nothing deleted and a test pinning the resulting boundary.

And MA23's own correction must be honoured, because it is the reason this is a proposal and not a chore. MA23 found, after the fact, that its move did **not** achieve one of its three stated motivations — the deploy image was unchanged, because `.dockerignore` already excludes `*.md`. So a documentation move buys **readability only** and must not be sold as anything else.

The concrete hazard, and it is specific. `valuation/web/research_record.py` — the public research page — **lists the pre-registration documents by globbing the filesystem**, not from a manifest. A move that does not update that glob **in the same commit** silently empties the public page's register list, and nothing would raise. That is the MA23 merge collision in a new place (two branches touching different files, a clean merge, and a broken import), and it is why this item carries a hazard rather than a checklist.

MB37 — Keep the `PROMPT_*` files, and say why

EV: LOW. Trials: 0.

Seven of them, each a commission that produced landed work. They are the evidence for the receipt convention MB29 proposes to make standard; deleting them would delete the case for it. They are also the only artifact in the tree that records *what was asked* as distinct from *what was found*, which is exactly the asymmetry the register exists to preserve.

MB38 — PIONEER, RANK 1 — HYPOTHESIS: publish the *denominator*, not the register — because the register already ships

EV: HIGH for a personal/portfolio site. Trials: 0. Cheapest first artifact: one paragraph and three numbers on a page that already exists.

Correcting the commission's premise, which is the useful part. The commission proposes "publishing the register method itself" as a novel artifact. **Half of it already ships.** `valuation/web/research_record.py` (item v4) serves `/research`, sourced from `RESEARCH_LOG.md` through **the same parse that produces the Deflated Sharpe's denominator** — deliberately, so the page cannot become a second version of the truth. Its rule is absolute and enforced by `withhold()` plus a test: **no performance figures, not results, not thresholds, not effect sizes**. The repository itself went **public on 2026-08-16** under MIT.

What exists in the world (searched). ClinicalTrials.gov (NLM/FDA, since 2000) and the AEA RCT Registry are the standard pre-registration registries for medicine and economics. **There is no standardised public pre-registration registry for quantitative-finance backtests.** Deflated-Sharpe and PBO calculators do exist as open source (pypbo; the `ml4t-diagnostic` package's `effective_number_of_trials()`; browser tools) — but those compute an effective number of **strategies tested**, inferred from a correlation structure. **None of them is a maintained, dated, adversarially-audited count of the trials one project actually ran.**

So the genuinely new artifact is the denominator, not the register. A public statement of the form: *this project has run 549 pre-registered trials — equity 234, options 300, infra 15 — the multiplicity-corrected critical value that count implies is $\sqrt{2 \cdot \ln 234} = 3.3031$, and the headline long-short statistic is 2.6199, which **does not clear it***. As far as this search establishes, no retail research record publishes its own honest denominator, derives its own hurdle from it, and reports its own headline failing that hurdle.

Why it is publishable when the alpha figure is not — the narrow, defensible opening. The withholding rule exists because of the public posture and because the Sharadar licence forbids commercial use of the data "*or any derivation*". **A trial count is not a derivation of vendor data**; it is a count of this project's own decisions. The hurdle is arithmetic on that count. The PASS/FAIL is a comparison of two numbers, one of which is already withheld — so the honest form publishes **N, the hurdle, and the verdict word**, and continues to withhold the statistic itself.

Cheapest first artifact: one paragraph and three numbers appended to `/research`. No new surface, no new data, no trials, one session.

Kill condition. If `withhold()` cannot be made to pass a trial count and a derived hurdle without also passing a performance figure, **do not ship** — the posture wins, and the failure is itself worth recording. Test it against the existing assertion that the rendered page contains no performance figure before writing any copy.

MB39 — PIONEER, RANK 2 — HYPOTHESIS: the disclosure-card class MA28 opened

EV: MEDIUM-HIGH. Trials: 0 for the strongest candidate. Cheapest first artifact: a second card on `accounting_risk.py`'s `pattern`.

What MA28-CARD-UI actually established is a product class, not a statistic. A card whose gate was a **crash rate rather than alpha**; whose thresholds this project **did not fit** (Beneish -1.78 and Altman 1.81 are published values); whose copy carries a **measured rule about its own instability** — *"the base rate is era-dependent, kept 0.3413% early against 1.3595% late, so quote the ratio and both rates, never the difference"*; and whose banned phrases are asserted against the **rendered payload** rather than the source. That combination — a published-threshold risk disclosure that states the conditions under which its own number misleads — is not a common product shape.

Three measured, un-carded candidates already in the record:

candidate	measured	trials to card it
S28's own distribution of the headline — the published +7.17%/yr alpha is the mean of 69 quarterly draws of which 20 are negative (28.99%) , median +1.41% against a mean +1.79%, worst -6.83% (2016-01-20)	already shipped in <code>construction.top_decile_alpha_distribution</code>	0
V6-B's survival gradient (MB10)	-14.287pp smallest quintile to -3.787pp megacap	0
P1S0's optionable partition (MB11)	worse early, better late, at all three horizons	0

The first is the strongest and nobody has proposed it. A card stating that the product's own headline was negative in nearly three quarters of every ten, with the worst quarter dated, is the most honest disclosure this project could ship — and S28 already computes it, ships it in the canonical artifact, and pinned it as **reporting-only** with a test that fails if any threshold ever branches on a distribution field. **The statistic is built, guarded and unused.**

Kill condition (inherited from `dip_posture.py`'s design): if the copy cannot pass a BANNED-phrase assertion against the **rendered** payload, do not ship it. Rendering is where copy leaks.

MB40 — PIONEER, RANK 3 — HYPOTHESIS: an open effective-n calculator, and why it ranks last

EV: LOW-MEDIUM. Trials: 0.

TIDEMARK's instrument — design effect measured against a **simulated no-predictability null carrying the series' own fitted persistence**, converted into required independent years at a multiplicity-corrected critical value — is, on this search, not packaged anywhere. The existing open tools compute effective **trials** (strategy clustering: `effective_rank`, Marchenko–Pastur, clustering); this computes effective **observations** (time-series clustering). The distinction is real and it is the one that bites retail backtesters, who overwhelmingly quote raw n.

Cheapest first artifact: one self-contained HTML page taking (`n`, `horizon`, `fitted rho`, `trial count`, `target IR`) and returning *"you need X independent years; you have Y."* TIDEMARK's dashboard is already exactly this shape — a single file, no external stylesheet, font or script, working with no network — so the template exists.

Why it ranks last, stated plainly. It is a tool, tools need users, and this project's audience is one person plus recruiters. Its genuine value is **internal** and is already captured by MB22, where the same arithmetic becomes a standard field on every register. Ship MB38 first; this only if the method is written up.

BUGS FOUND (RUN_RULES Part A rule 3 — including outside this lane)

MB42 — A gate suite is GREEN in CI and RED on the only machine that owns the data, and the cause is a path separator

Lane: options bot / data miner. Trials: 0. Cost: one literal.

The audit's sanity check ran every suite by exit code, as RUN_RULES 0.1 requires. **115 suites, 114 passed, 1 failed** — tests/test_o21d2_alternative_pnl.py, at test_the_real_harvest_freeze_resolves_when_mounted (20 of its 21 assertions pass).

Diagnosed, not guessed. The test asserts four things about the resolved harvest provenance. Three pass — pinned is True, hash_mismatches_at_copy is 0, manifest_sha256 is 64 chars. The fourth is:

```
assert prov["frozen_from"] == "D:/thetadate/chains"      # forward slashes
actual: 'D:\\thetadate\\chains'                          # backslashes
```

It is a path-separator literal, not a data problem. The freeze is intact:

```
D:\thetadate\freeze_rawpull_2026-08-18, pinned true, zero hash mismatches, manifest sha256
ee6d38e5ff58...f000.
```

The part that makes it worth reporting rather than just fixing. The test opens with a mount guard — *"if not os.path.isdir(CS.harvest_root()): print SKIP; return"*. The CI runner is Linux and has no D: drive, so **the assertion never executes there and the suite exits 0**. The auto-land Action is therefore green while the same suite is red on the only machine where the data it guards actually exists. **A guard that can only fire where nobody runs it is not measuring anything** — the same family as MA5's source sweep firing on its own documentation and MA23's stale-path guard being blind to the import syntax the codebase writes.

Fix: compare normalised paths (`os.path.normcase(os.path.normpath(...))` on both sides), so the assertion means the same thing on both platforms. **Kill condition:** if normalising makes the test pass on Windows *and* it still skips on Linux, add a fixture so the comparison runs on both — a guard whose only real execution is skipped is the defect, not the separator.

Not fixed here: this audit is read-only, and it is not this lane's file.

QUESTIONS FOR DON — batched, none blocking

1. **MB21 is the audit's top item and it may end with copy being withdrawn.** If the persistence-preserving null shows S22's long-horizon result inside it, the hot-list card's *"still ahead by about 5.1% annualized two years later"* comes off. Do you want that pre-committed in the register before the number is read (my recommendation, and the project's own discipline), or do you want to see the number first and decide?
2. **MB14 spends 1 equity trial on a probable "cannot tell".** I recommend it — it converts the project's largest open finding into a dated, powered refusal and makes re-litigation costly. Say no and it stays open indefinitely, which is also a defensible choice.
3. **MB2: 16 trials for the full DTE × delta grid, or 1 for the 60–90 DTE cell alone, or zero.** I recommend zero and explain why; you have pre-committed interest in 60–90 DTE, so this is genuinely your call rather than mine.
4. **MB3 needs no permission but produces a number you may not like:** the account equity at which an earnings-spanning book survives 011's ruin arithmetic. If it comes back above \$250,000, do you want it closed permanently or held open?
5. **MB25 (the TIDEMARK page):** linked page at /tidemark, or a tab? I recommend a linked page and give the reason (two verdict grammars, two denominators). Also — are you content that the page leads with TIDEMARK's refusal sentence rather than its numbers? That is the honest ordering and it is a product choice.
6. **MB28 is Cowork's, not mine:** registering a Windows scheduled task to run `scripts/checkout_drift.py --json` daily. It cannot be done in CI (MA11 refuses .github/ edits, and it measures a local checkout). Shall it be routed?
7. **MB38 publishes N, the derived hurdle, and the word FAIL — not the statistic.** That is a posture decision, not a research one. Comfortable?
8. **MB36 moves 116 markdown files into register/ and handoff/.** It buys readability only (not image size — MA23 proved that) and it carries one real hazard (`research_record.py` globs the filesystem). Worth the commit, or leave the root as-is?

THE ANSWER TO THE QUESTION THIS COMMISSION ACTUALLY ASKS

After ~550 trials across two books, **the next real edge most plausibly lives in the instruments rather than in another hypothesis** — and the strongest single candidate is not a signal at all but a null: the composite's own

persistence is 0.5802 at one quarter and still 0.4099 at two years, while the placebo it is judged against has exactly zero, so S22's term-structure claim — the one result this project chose to put on the product's front card — has never faced the artifact that Boudoukh–Richardson–Whitelaw say produces its exact shape. Behind it, in descending plausibility: the 2.7 million alternative contracts, which permit the one decomposition (R2 as timing versus selection) that closes a family rather than opening one; the tick tape's 14-venue exchange field, unread by all five 014 features and all of 010/018, and the literature's own retail identifier; and the banked reverse-DCF panel's point-in-time implied-growth gap, at 100% coverage over 69 dates and never scored. **Nobody should ever look again at:** regime conditioning of the equity composite, where the arithmetic is not close — 34.2 years of data per side against 17.3 in total, and the required gap exceeds the entire published alpha at any critical value; the options-expression family, closed by P1S0 at its power anchor and priced by DEEPITM-FIN at 702 bps/yr all-in against margin cards at 150 and 420; short vol, closed by 09 and re-killed by V6-0PT's mechanism; and any register whose motivation is "*structurally orthogonal to the incumbents*", which has now failed three times on this panel with R2 between 0.027 and 0.088 and nothing to show for it. **And the single cheapest experiment that could genuinely surprise us is MB21's** — permute each name's whole signal time series instead of permuting within dates, rebuild S22's per-horizon floor against a null that finally remembers, and look at H=504. It costs one infrastructure trial, needs no new data, runs in a session, and it is the only experiment in this document whose most likely outcome is that the project has to take something off the product — which, on a record that is overwhelmingly rejections, would be the most informative result available.

Sources cited from outside the corpus

- • Blitz, D. — *The Quant Crisis of 2018–2020: Cornered by Big Growth* (Robeco, 2021). robeco.com · PDF. *Replication status: the episode's dating and character are corroborated across independent practitioner research (Man Group, First Sentier); no causal mechanism is claimed here.*
- • Boudoukh, Richardson & Whitelaw — long-horizon R² rising mechanically under the null (RFS 2008); and Boudoukh, Israel & Richardson (JFE 2022) on bias-adjusting long-horizon regressions. *Cited via TIDEMARK's CHARTER.md §3.2, which reproduces the check on its own data.*
- • Bryzgalova, Pavlova & Sikorskaya — *Retail Trading in Options and the Rise of the Big Three Wholesalers*, **Journal of Finance** 78(6), 2023, 3465–3514. Wiley · SSRN. *Replication status: published in the JF, widely cited. The wholesaler flag it uses is NOT available from this project's vendor; any venue-based classifier here is a proxy. Extended by Bogousslavsky & Muravyev, An Anatomy of Retail Option Trading (PDF).*
- • Easley, López de Prado & O'Hara — VPIN / From PIN to VPIN (PDF); **contested by** Andersen & Bondarenko, *Assessing Measures of Order Flow Toxicity* (SSRN, PDF) and *VPIN and the Flash Crash* (SSRN); response in *Review of Finance* 19(1). *Replication status: DISPUTED, and the disputed component is the Bulk Volume classifier — which is why only the quote-classified version is proposed (MB16).*
- • Gao, Han, Li & Zhou — *Market Intraday Momentum*, **JFE** 2018 (ScienceDirect, SSRN). *Marked NOT-TESTABLE-HERE in MB17: it is an index/ETF half-hour phenomenon and this project owns no index chain and no equity tape.*
- • Pre-registration registries surveyed for MB38: ClinicalTrials.gov (NLM/FDA) and the AEA RCT Registry. Open backtest-diagnostic tooling surveyed: pypbo, ml4t-diagnostic, quant4free tools, quantstrat::SharpeRatio.deflated. *None is a maintained count of one project's own trials.*